

AE 520 Fall 2011 Compressible Flow

Time, place: MWF 2.30 pm - 3:30 pm, 1024 FXB
Instructor: Prof. Bram van Leer
Office hours: MW 3:30 - 5:00 pm, 3000 FXB
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TA: Daniel Zaide (2037 FXB)
Office hours: Th 1:00 - 2:15 pm, F 3:30 - 5:00 pm,
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Lecture notes: on CTools website under Resources
Textbook: the books listed below cover part of the material.

Ya. B. Zel'dovich and Yu. P. Raizer,
"Physics of Shock Waves and High-Temperature Hydrodynamic Phenomena,"
Volume 1, Chapter 1, Academic Press 1966,
compact introduction to 1-D gas dynamics,
reprinted as course pack 1991-1997;
may be posted on CTools website.

H. W. Liepmann and A. Roshko,
"Elements of Gasdynamics," Wiley, 1957,
venerable text, used for this course up to 1990.

R. Courant and K. O. Friedrichs,
"Supersonic Flow and Shock Waves," Interscience, 1948,
classic reference book.

G. B. Whitham, "Linear and Nonlinear Waves," Wiley, 1974,
modern reference on waves; includes traffic flow.

K. Masatsuka, "I do like CFD, Vol. 1, 2009,"
Authored and published by UM graduate Hiroaki Nishikawa,
this covers "Governing Equations and Exact Solutions."
Get it from amazon.com for \$9.99 plus shipping!

Grading

Problem sets: 5	20%
One-hour exam #1 (W Oct 19)	15%
One-hour exam #2 (F Nov 18)	15%
Computer problem (due W Nov 24)	20%
Final exam (F Dec 16, 1:30-3:30)	30%

Outline

Date	Subject/Event	Textbook sections
	1. Introduction	
Sep 07	Thermodynamic properties of gases, equation of state, entropy, sound speed	ZR 1, 3; LR 1.3-4, 1.6, 1.7, 1.9-11, 1.13, 2.8, 3.6
09	Integral/differential flow equations Jump equations, entropy condition I	ZR 1; LR 7.1-8 ZR 13; LR 1.9, 2.12
12	Moving shocks, x-t diagram, different frames	ZR 13; LR 3.2
14	Relation to steady supersonic flow, entropy condition II	LR 4.6
16	Entropy conditions III, finite-volume method Exact solutions <i>vs.</i> CFD	- -
	2. 1-D unsteady flow: linearized	
19	Acoustic equations, d'Alembert's solution	ZR 3; LR 3.4-5
21	Forward/backward Method Of Characteristics Linear jump equations	ZR 3; LR 3.7 -
23	Piston problem, simple waves	-
26	Riemann problem Problem Set #1 due	LR 3.8
	3. 1-D unsteady flow: fully nonlinear	
28	Characteristic equations, simple waves	ZR 5-8
30	Piston problem (expansion), MOC	ZR 10-11, 6; LR 3.11
Oct 03	Particle path in simple wave Problem Set #2 due	-
05	Riemann problem Linear vs. nonlinear Riemann problem	ZR 17; LR 3.12 Z 18
07	u-p diagram, Poisson/Hugoniot curves	ZR 14-16, 24
10	Iterative solution of Riemann problem	-
12	Godunov's finite-volume method	-
14	Piston problem (compression)	ZR 24
17	Fall Recess	
19	One-hour exam #1	
21	Wave interactions, analogue in 2-D steady supersonic flow	ZR 16, 24

4. Burgers' equation		
Oct 24	Inviscid Burgers equation for simple waves	-
	Jump equation, shock formation,	-
	Entropy condition II	-
26	Wave interactions	-
	Problem Set #3 due	
28	Wave interactions	-
31	Viscous Burgers equation, entropy condition III	-
Nov 02	Traffic equation	-
04	Buffer time	
5. 2-D flow equations		
07	Unsteady flow equations	-
	Characteristic equations via linear algebra (1-D)	-
	Problem Set #4 due	
09	More linear algebra (2-D), hyperbolicity	-
	Two versions of Huygens principle	-
6. 2-D steady supersonic flow		
11	Steady supersonic flow, characteristics	LR 12.1-3
	Time- and space-like coordinates	
14	Natural coordinates	-
16	Riemann invariants, Prandtl-Meyer function	LR 4.10, 4.20
18	One-hour exam #2	
21	Crocco's theorem	
23	Method of characteristics, simple waves, nozzle design	LR 12.4, 12.7, 12.11
	Computer Problem due	
25	Thanksgiving break	
7. Burgers for steady supersonic flow		
28	Inviscid Burgers equation for simple waves, rotationally invariant form	-
30	Wave interactions	-
Dec 02	Flow over diamond-shaped airfoil	LR 4.16, 8.7
05	2-D small-disturbance theory	LR 8.1-3
	Transonic small-disturbance equation	LR 10.3
07	Subsonic/supersonic flow over wavy wall	LR 8.5-6
09	Buffer time	
	Problem Set #5 due	
12	About the final exam	
16	Final exam (1:30 - 3:30 pm)	

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